

Maven Market 1997 and 1998 Market Analytics Report

At Maven Market, our mission is to continually grow while serving our customers to the best of our abilities. Over the years 1997 and 1998, we have experienced significant growth in our three regions: USA, Canada, and Mexico, with our numbers showing no signs of slowing down.

This report provides a detailed analysis of our progress over these years, prepared by the Data Analytics team using information from our Data Warehouse and Data Engineering team.

Assumptions During ETL Process

1. **Maven Market Calendar:** Assumed to be accurate; no further analysis conducted.
2. **Maven Market Customers:** Assumed to be accurate; no further analysis conducted.
3. **Maven Market Products:**
 - The "recyclable" column had empty cells. The assumption is that a value of "1" indicates the product is recyclable, as only "1" and empty values were present. Empty cells were assigned a value of "0" to indicate non-recyclable.
 - The "low_fat" column had similar issues and was treated the same way. These assumptions will be revisited if these columns are used in calculations.
4. **Maven Market Regions:** Assumed to be accurate; no further analysis conducted.
5. **MavenMarket_Returns_1997-1998:** Assumed to be accurate; no further analysis conducted.
6. **MavenMarket_Stores:** Assumed to be accurate; no further analysis conducted.
7. **MavenMarket_Transactions_1997:** Assumed to be accurate; no further analysis conducted.
8. **MavenMarket_Transactions_1998:** Assumed to be accurate; no further analysis conducted.
 - Transactions from 1997 and 1998 were combined using append queries to create a new table named "MavenMarket_Transactions_1997-1998".

This report aims to provide our stakeholders with a comprehensive overview of our data and insights derived from it.

Summary of Results

Table 1: Summary of Sales

Store_Country	Year	Yearly Revenue	Best Month Sale	Best Month	Worst Month Sale	Worst Month	Average Monthly Sale
USA	1997	\$565,238.13	\$56,965.64	December	\$42,342.27	October	\$47,103.18
USA	1998	\$612,718.31	\$66,060.29	November	\$45,692.89	October	\$51,059.86
Mexico	1998	\$478,915.66	\$48,622.07	December	\$37,462.18	November	\$39,909.64
Canada	1998	\$107,674.34	\$10,265.37	November	\$7,144.80	January	\$8,972.86

The table summarizing sales data for different Mavin Market stores across various countries and years. It includes information on yearly revenue, best and worst month sales, and average monthly sales.

General Trends

- **Yearly Revenue:** Increased for the USA between 1997 and 1998.
- **Best Month:** Varied by country and year.
- **Worst Month:** October seems to be a consistent underperformer for the USA.



Figure 1: Market Performance

The image presents the performance of Mavin Market stores. This image offers a high-level summary of overall business performance. The metrics displayed in the image are likely derived from the aggregated data in the table.

Insights

- **Positive Performance:** The high revenue and number of orders suggest strong sales activity.
- **Profitability:** The profit margin, implied by the profit figure relative to revenue, seems reasonable. However, a more detailed profit and loss analysis would be necessary to assess profitability accurately.
- **Low Return Rate:** The extremely low return rate indicates high customer satisfaction or effective product quality control.

Table 2:Summary of Sales by Region

Total Sales by Reg	
Store_Country	Total_Region_Sales
USA	1,177,956.44
Mexico	478,915.66
Canada	107,674.34
Total	1,764,546.44

This image offers a basic overview of sales distribution across different countries. The image displays a table that contains two columns: This image provides a more detailed breakdown of the total sales figure presented in the previous image. It specifies the contribution of each country to the overall sales.

Potential Insights:

- **Dominant Market:** The USA has the highest sales figure, indicating it's the primary market for the business.
- **Market Share:** By calculating the percentage of total sales for each country, you can determine the market share of each region.
- **Growth Opportunities:** Comparing sales figures over time can identify regions with growth potential

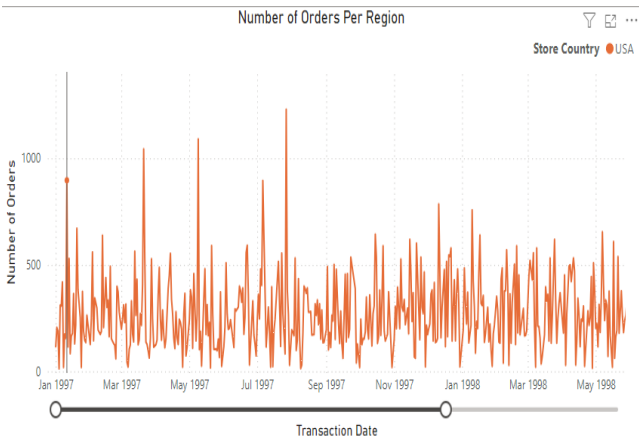


Figure 2: Number of orders (USA)

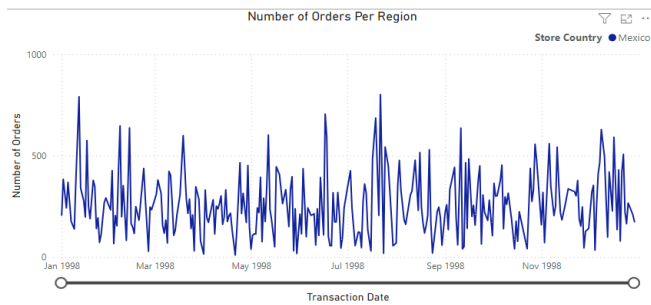


Figure 3: Number of orders (Mexico)

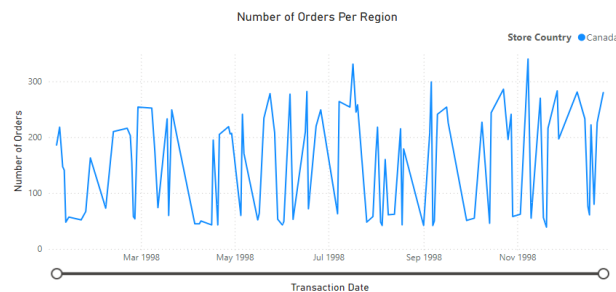


Figure 4: Number of orders (Canada)

These charts offer a visual representation of order volume over time for the USA region, Mexico and Canada region.

Observations:

The number of orders exhibits significant fluctuations over time, with peaks and valleys. There seems to be a potential seasonal pattern in the data, with higher order numbers during certain periods of the year. Some data points deviate significantly from the overall trend, indicating potential anomalies or special events.

Potential Insights: Analysing the peaks and valleys can help identify factors influencing sales, such as promotional activities, holidays, or economic conditions. Examining the overall trend can reveal growth or decline in order volume over time. Comparing this chart with similar charts for other regions can highlight regional differences in sales patterns.

Comparisons:

Compared to the "USA" region, the Mexico data exhibits higher volatility. While both regions show potential seasonality, the pattern in Mexico might be less pronounced or different in timing.

Order Volume: Canada has consistently lower order volumes compared to the USA and Mexico.

Volatility: The Canadian data exhibits less volatility compared to the other two regions.

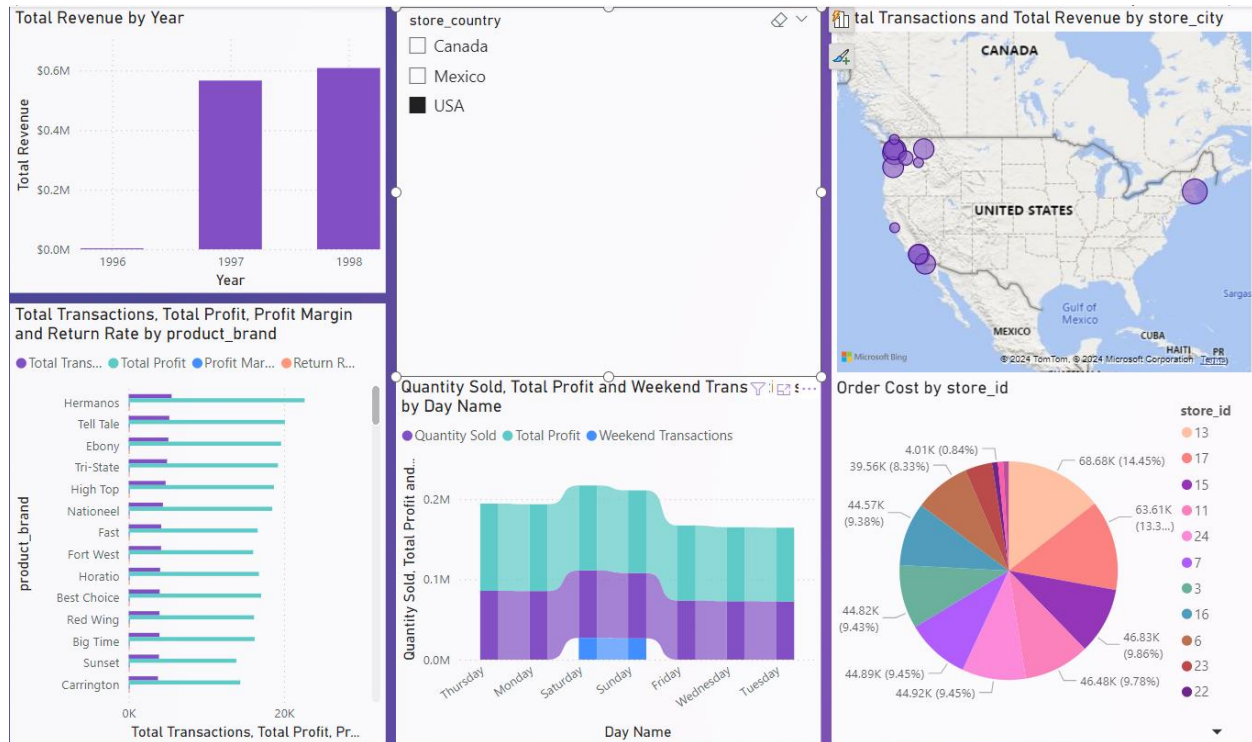


Figure 5: Summary of individual product performance per region

These images provide a high-level overview of dashboard of Mavin Market business performance across various dimensions.

Breakdown of Visualizations:

- Total Revenue by Year:**
- Total Transactions and Total Revenue by Store City:** A map of a region, with multiple cities marked. Each city has a circle representing total transactions and total revenue. The size of the circle correlates with the magnitude of transactions.
- Total Transactions, Total Profit, Profit Margin, and Return Rate by Product Brand:** A bar chart comparing multiple product brands based on four metrics: total transactions, total profit, profit margin, and return rate. Each brand is represented by a group of four bars, one for each metric. These show the brands which are most patronized by clients.
- Quantity Sold, Total Profit and Weekend Transactions by Day Name:** A stacked bar chart showing quantity sold, total profit, and weekend transactions for each day of the week. The height of each bar represents the total value for that day, with different colours indicating what days transactions are
- Order Cost by Store ID:** A pie chart displaying the distribution of order costs across two store IDs: 19 and 20. The size of each slice represents the proportion of total order cost for that store ID.

Potential Insights:

- **Revenue:** The total revenue for 1998 can be used as a baseline for performance evaluation.
- **Geographic Distribution:** The map provides insights into the geographic distribution of sales and potential areas for growth.
- **Product Performance:** The product brand comparison helps identify top-performing and underperforming brands based on different metrics.
- **Sales Patterns:** The daily sales pattern analysis can reveal peak sales days and opportunities for optimization.
- **Cost Analysis:** The order cost distribution by store ID can help identify cost-efficiency differences between stores.

Further Analysis:

- **Time Series Analysis:** Incorporating data for earlier years would provide a longer-term perspective on revenue growth.
- **Customer Segmentation:** Analyzing customer data can identify high-value customers and tailor marketing efforts accordingly.
- **Competitive Analysis:** Comparing performance metrics with industry benchmarks can assess competitive positioning.
- **Drill-Down Capabilities:** Providing the ability to explore data at a more granular level would enhance insights.

Recommendations

- **Revenue:** Consistent growth is observed across multiple images, indicating overall business expansion.
- **Geographic Performance:** The USA is consistently the top-performing region in terms of sales and transactions.
- **Product Performance:** While specific product data is limited, the comparison of product brands in Dashboard 1 provides initial insights into performance.
- **Sales Patterns:** The analysis of sales by day of the week reveals potential opportunities for optimization.
- **Operational Efficiency:** The distribution of order costs by store ID highlights areas for potential cost reduction.

ANALYSIS

- The company has experienced significant revenue growth over the analyzed period.
- The USA is the primary market, but further analysis of Mexico and Canada is warranted.
- Product performance analysis should be deepened to identify top-performing and underperforming products.

- Optimizing sales efforts based on peak sales days can potentially increase revenue.
- Cost reduction initiatives should focus on high-cost stores.